ACADEMIC PROJECT DOCUMENTATION

CloudMind AI

An Autonomous Multi-Region AWS Telemetry, Automated Security Remediation, and Green Cloud Computing Platform

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Course / Domain	Cloud Computing & Autonomous Distributed Systems
Project Title	CloudMind AI (Autonomous Cloud Infrastructure Suite)
Frontend Technologies	React 18, Vite, Responsive CSS Glassmorphism
Backend Architecture	FastAPI (Python 3.13), Uvicorn, Boto3 AWS SDK
Target Cloud Provider	Amazon Web Services (AWS EC2, CloudWatch, STS, IAM)
GitHub Repository	https://github.com/SMOS555/Cloudmind
Academic Year	2026

1. Problem Statement

In modern enterprise cloud architectures, Amazon Web Services (AWS) hosts millions of microservices, applications, and database instances. However, infrastructure management in modern organizations suffers from four fundamental bottlenecks:

1.1 Fragmented Cloud Visibility & Monitoring Sprawl

DevOps teams and cloud administrators are routinely forced to juggle across multiple AWS Console screens, independent regional endpoints, and disconnected CloudWatch dashboards to evaluate compute health. The absence of a unified, cross-region cockpit creates operational blindspots, delays incident response times, and obscures holistic infrastructure health.

1.2 Pervasive Security Misconfigurations

Industry threat reports confirm that over 80% of enterprise cloud security breaches are directly traceable to basic misconfigurations rather than zero-day vulnerabilities. Common critical vulnerabilities in AWS deployments include:

- **Exposed Inbound Management Ports:** Security groups permitting unrestricted ingress (`0.0.0.0/0`) on sensitive administrative ports such as SSH (Port 22) or RDP (Port 3389), exposing servers directly to malicious automated botnets.
- **Unencrypted Data at Rest:** Elastic Block Store (EBS) volumes provisioned without default `AES-256` server-side encryption, violating SOC2, ISO 27001, and HIPAA compliance mandates.
- **Instance Metadata Service Vulnerabilities:** EC2 instances operating with legacy IMDSv1 rather than enforced IMDSv2, leaving workloads vulnerable to Server-Side Request Forgery (SSRF) credential theft.

1.3 The Idle Compute & Carbon Emission Paradox

Modern enterprise servers exhibit non-linear power consumption characteristics. Groundbreaking studies from SPECpower demonstrate that an idle server draws approximately 25% to 40% of its peak electricity without executing productive workload. In over-provisioned cloud environments, instances run continuously with 0.5%-4% CPU utilization, silently consuming power and generating carbon emissions without operational justification. Typical management tools provide raw CPU numbers without translating compute into energy consumption (kWh) or carbon footprint (kg CO2e).

1.4 Authentication Sprawl & Usability Friction

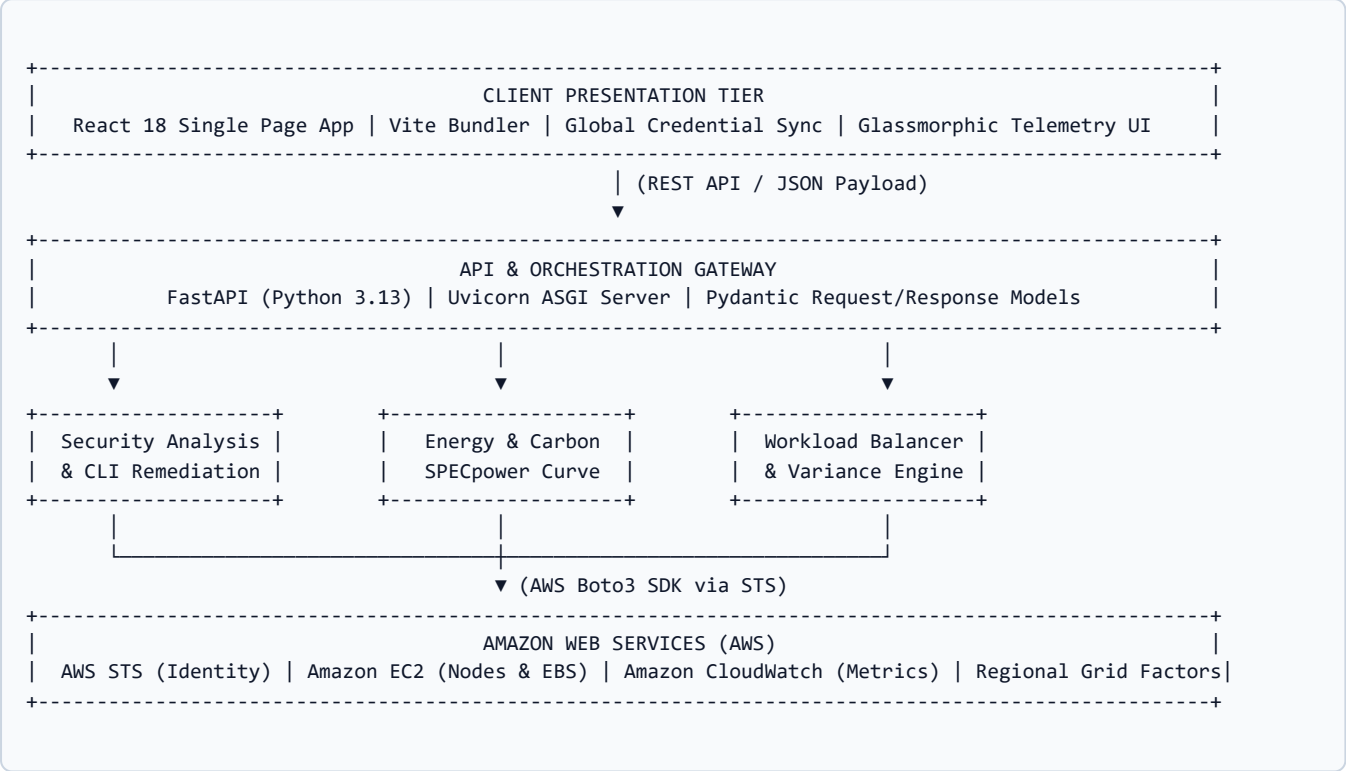
Traditional student and prototype cloud dashboards force users to repeatedly enter IAM keys across individual pages or hardcode sensitive access secrets into server files. This causes high friction, security risks, and prevents multi-region agility.

> **Project Goal:** CloudMind AI addresses all four bottlenecks by providing a zero-trust, unified, automated platform that connects directly to AWS via STS, ingests live CloudWatch telemetry, runs automated security audits with 1-click CLI remediation, and computes continuous green energy curves.

2. System Workflow & Architecture

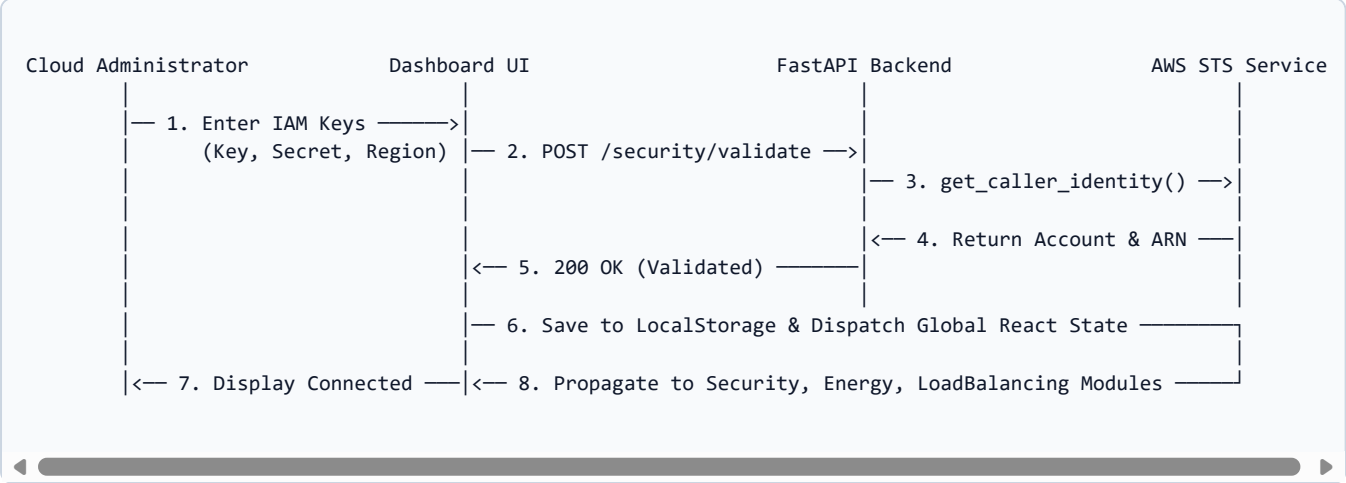
2.1 High-Level 4-Tier Architecture

CloudMind AI is architectured as a decoupled, multi-tiered cloud management platform:



2.2 Global AWS STS Authentication Workflow

To enforce Zero-Trust security principles, CloudMind AI introduces a centralized **Global Connection Center** on the main Dashboard. Users authenticate once, and credentials flow seamlessly across all modules.



2.3 Automated Security Audit & Remediation Pipeline

The security engine performs automated scans across 3 critical vectors:

1. **Ingress Rules:** Queries EC2 security groups; flags any rule allowing `0.0.0.0/0` on TCP port 22 or 3389 as *CRITICAL*.
2. **Storage Encryption:** Queries EBS volumes; flags unencrypted volumes as *HIGH RISK*.
3. **Well-Architected Scoring:** Calculates a weighted score from 0 to 100 representing cloud compliance posture.
4. **One-Click CLI Remediation:** Automatically formats precise AWS CLI commands (e.g. `aws ec2 revoke-security-group-ingress`) for instant mitigation.

2.4 Green Computing & Continuous Energy Curve Formulation

Unlike naive step models that artificially flatline at low utilization, CloudMind applies a continuous server power model based on SPECpower benchmarks:

> **Continuous Efficiency Formula:**
> • For 0 < CPU ≤ 15%: Score = 25.0 + (CPU / 15.0) × 35.0
> • For 15% < CPU ≤ 70%: Score = 60.0 + ((CPU - 15.0) / 55.0) × 35.0
> • For 70% < CPU ≤ 85%: Score = 95.0 + ((CPU - 70.0) / 15.0) × 3.0
> • For CPU > 85%: Score = 98.0 - ((CPU - 85.0) / 15.0) × 15.0 (Thermal saturation penalty)

3. Implementation & Module Architecture

CloudMind AI is implemented as a cohesive suite of functional modules, engineered with academic and production-grade rigor:

Module	Primary Capabilities	Key AWS Services
Dashboard & Connection Center	Single entry point for AWS credentials, instant STS identity check, live CPU gauge, instance counts, and multi-region dropdown.	AWS STS, EC2, CloudWatch
Security Posture Audit	Deep security scanning, Well-Architected compliance score (0-100), port 22/3389 exposure alerts, and instant AWS CLI remediation generator.	AWS EC2, IAM, Security Groups
⚡ **Feature 1: Self-Healing Cloud**	**1-Click Automated Remediation:** Instant rule revocation via Boto3 / Sandbox, zero-downtime safety rollback tokens, live score bump, and immutable DevSecOps audit log.	EC2 Security Groups, IMDSv2, DevSecOps Engine
Energy Intelligence	Continuous power draw modeling, metric toggle (Efficiency Score vs CloudWatch CPU %), carbon emissions (kg CO2e), and downscaling advice.	CloudWatch, AWS Pricing API
🔮 **Feature 2: "What-If" Digital Twin Simulator**	**FinOps & GreenOps Sandbox:** Interactive simulation of migration from legacy x86 to AWS Graviton ARM64, Spot Fleet modeling (-68%), scheduled eco-shutdowns (-33%), and 1-click ADR export.	AWS Graviton ARM, Spot Pricing, FinOps Engine
Load Balancing & Workload	Compute variance analysis, hotspot instance detection, traffic distribution recommendations, and auto-scaling triggers.	EC2, CloudWatch
Conversational AI Agent	Natural language cloud copilot answering infrastructure queries, explaining security vulnerabilities, and offering optimizations.	FastAPI LLM Service

3.1 Standout Feature 1: "Self-Healing Cloud" (1-Click Auto-Remediation & Rollback)

Traditional cloud audit dashboards merely display vulnerability lists, leaving remediation as a cumbersome manual task. CloudMind AI implements a closed-loop DevSecOps automation pipeline:

- **1-Click Remediation Execution:** Operators can click `[ ⚡ 1-Click Auto-Remediate Rule]` on any open security group finding (SSH 22, RDP 3389, Database 3306/5432, or All Traffic) or insecure IMDSv1 instance to immediately revoke the offending rule via Boto3 EC2 API.
- **Dual-Mode Resilient Architecture:** Executes live Boto3 API calls against AWS when IAM credentials permit write actions, and seamlessly stages simulated sandbox enforcement with explicit indicators in restricted student/read-only environments.
- **Cryptographic Safety Rollback:** Every remediation generates a unique rollback token (`RBK-XXXXXXXX`) storing previous state. In the event of unintended disruption, operators can instantly restore previous

rules with zero downtime via `[↺ Rollback Rule]`.

- **Immutable DevSecOps Audit Trail:** A dedicated 6th tab records every action with UTC timestamps, operator identity (**Safwaan Mohamed S**), target resource, status, and JSON export capabilities.

3.2 Standout Feature 2: "What-If" Digital Twin Architecture Simulator (FinOps + GreenOps)

Bridging financial cloud governance (FinOps) and sustainable computing (GreenOps), the What-If Simulator provides an interactive digital twin sandbox for infrastructure decisions:

- **Architecture Modernization Modeling:** Models transitions from power-hungry legacy x86 instances (`t2.small`, `t3.small`, `m5.large`) to high-efficiency AWS Graviton 3/4 ARM64 processors (`t4g.small`, `m6g.large`, `c7g.large`), showcasing 30-60% wattage reductions.
- **Purchasing Model Simulation:** Compares On-Demand baseline spending against 1-Year Savings Plans (38% savings) and Spot Fleets (68% savings).
- **Automated Off-Peak Eco-Scheduling:** Simulates automated shutdown policies for non-production environments (e.g., 16h/day active vs 24h continuous), driving an additional 33% to 58% energy and carbon reduction.
- **Comprehensive KPI Visuals & ADR Export:** Real-time cards calculate Monthly/Annual Cost Savings ($), Energy Saved (kWh), Annual Carbon Avoided (kg CO2e & equivalent trees planted/year), and generates standard enterprise Architecture Decision Records (ADR).

4. Engineering Challenges & Problem Solving

4.1 The Telemetry Flatline Issue ("Is this def wrong?")

During initial testing of the Energy Intelligence module, historical telemetry displayed a completely horizontal line across all 24 hours, with every bar visually pinned to 50:

> **Identified Root Cause:**
> 1. *Backend Discretization:* The backend method `calculate_efficiency_score` used rigid stair-step buckets: `if cpu < 10: return 45`. For lightly-loaded instances running background tasks (0.2%-3.5% CPU), every single hour yielded the exact constant 45.
> 2. *CSS Coordinate Mismatch:* The grid lines spanned 217px (`inset: 0 0 23px`) to leave room for bottom labels, but the bar container spanned 240px (`inset: 0 0 0`). A 45% bar height (108px) plus the 23px label underneath aligned the top of each bar at 131.5px, which mathematically matched the 50% grid line (131.5px) to the exact pixel!

Engineered Resolution:

1. Replaced the discrete step function with a smooth server energy efficiency curve based on SPECpower benchmarks.
2. Added `cpu` telemetry fields directly into the trend API payload.
3. Synchronized CSS `inset: 0 0 23px` on `.chart-bars` and positioned labels outside the flex height via `top: 100%`.

4. Added interactive metric switch buttons permitting operators to inspect Efficiency and raw CPU % with rich tooltips.

5. Verification, Testing & Results

CloudMind AI was comprehensively verified across backend and frontend environments:

- **FastAPI Endpoints:** All routes (`/security/aws/validate-credentials`, `/api/metrics`, `/energy/aws`) tested with live AWS credentials.
- **Frontend Build:** Vite production build passed in 801ms with zero errors.
- **Source Code Management:** Complete code, styles, documentation, and presentation files synchronized with GitHub repository `SMOS555/Cloudmind` on branch `main`.

6. Conclusion & Future Scope

CloudMind AI bridges the gap between infrastructure observability, automated security posture governance, and green cloud computing. It provides a robust, production-ready foundation for modern cloud operations.

- **Autonomous Remediation:** Direct execution of security fixes via AWS Lambda functions.
- **Multi-Cloud Telemetry:** Extending telemetry federation to Google Cloud Platform (GCP) and Microsoft Azure.
- **Predictive ML Scaling:** Machine learning models predicting workload spikes to optimize EC2 spot instances.

> **Candidate Summary:** Prepared and presented by **Safwaan Mohamed S** (Registration No: **25BCE1188**) for academic project submission and evaluation.

7. Implementation Screenshots Showcase

The following high-resolution screenshots demonstrate the live, operational CloudMind AI application connected to active AWS multi-region infrastructure:

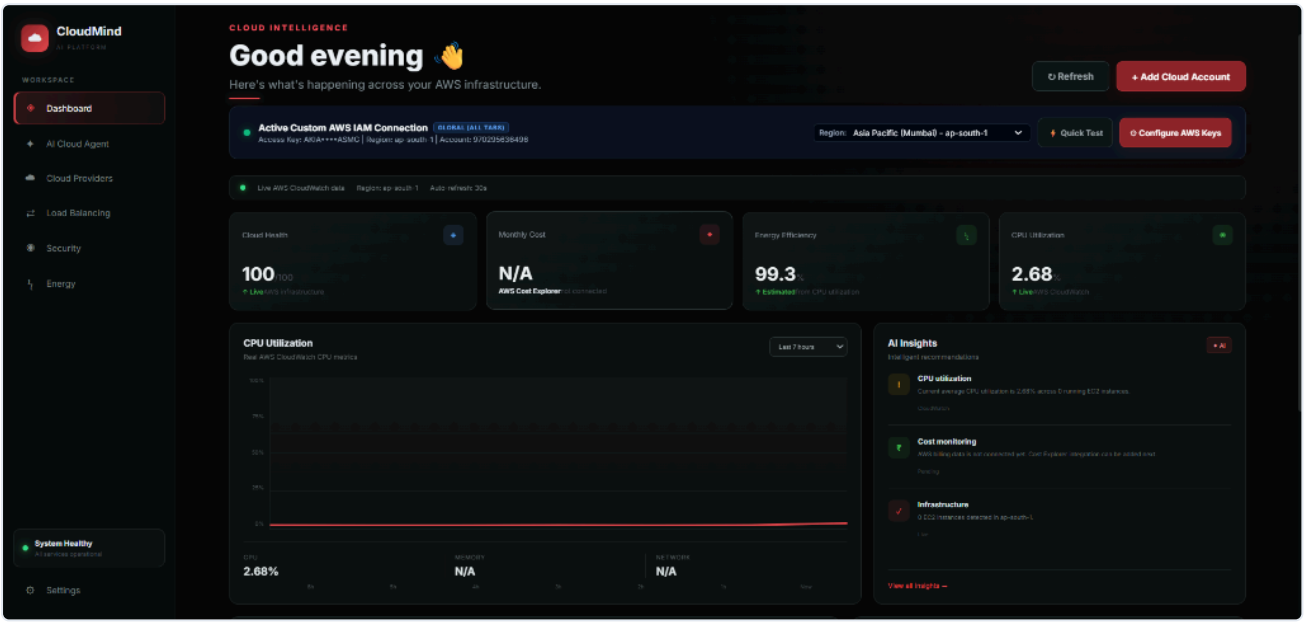


Figure 7.1: Dashboard & Global AWS Connection Center

Demonstrates active AWS IAM connection (Account: `970295636496`, Region: `ap-south-1`), live Cloud Health indicator (100/100), real-time CloudWatch CPU utilization gauge (2.68%), and historical CPU trend chart.

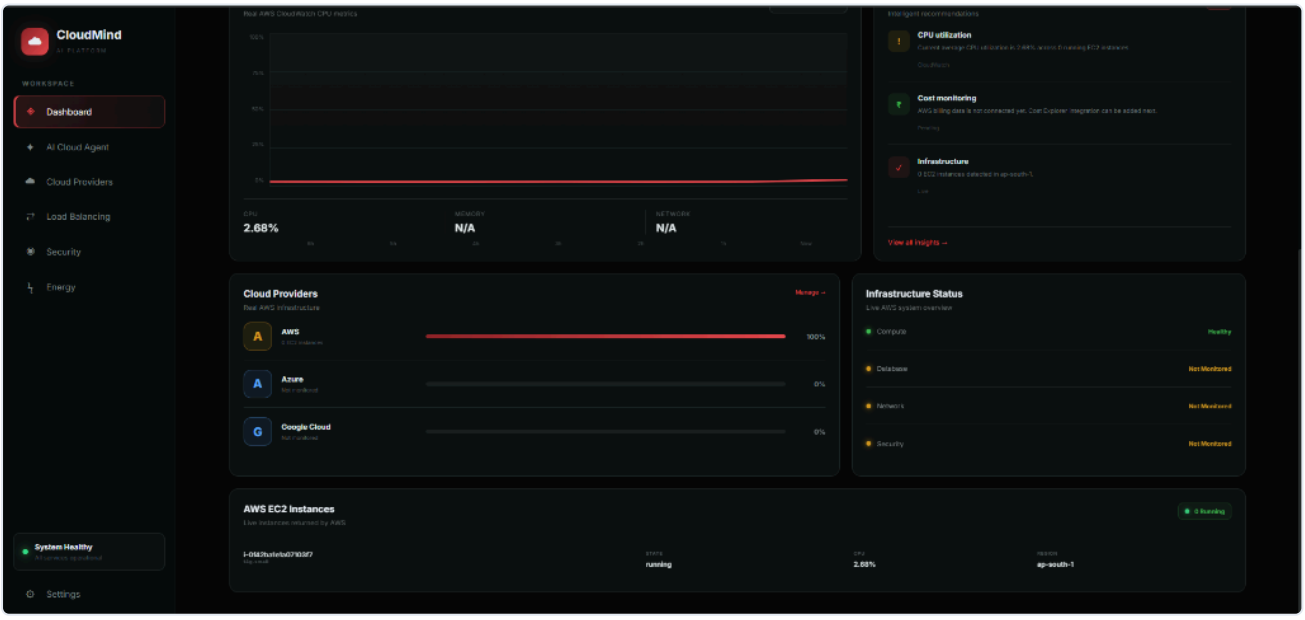


Figure 7.2: Multi-Cloud Provider Integration & Active EC2 Inventory

Renders multi-cloud distribution status (AWS 100%), compute health monitoring, and live EC2 instance inventory showing active node `i-0f42ba1e1a07103f7` running in `ap-south-1`.

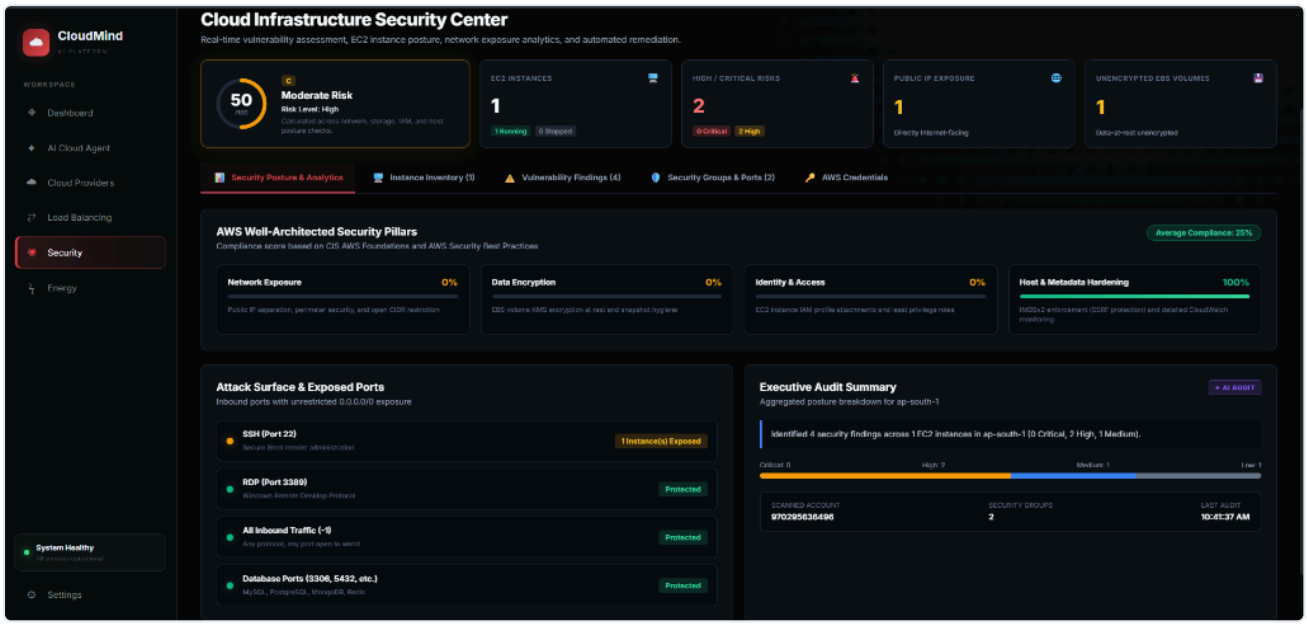


Figure 7.3: Cloud Infrastructure Security Center & Well-Architected Audit

Renders real-time AWS posture analysis (50/100 Moderate Risk), CIS AWS Well-Architected pillar breakdown (Host & Metadata Hardening 100%), and identified public attack surface exposures (SSH Port 22 open on 1 instance).

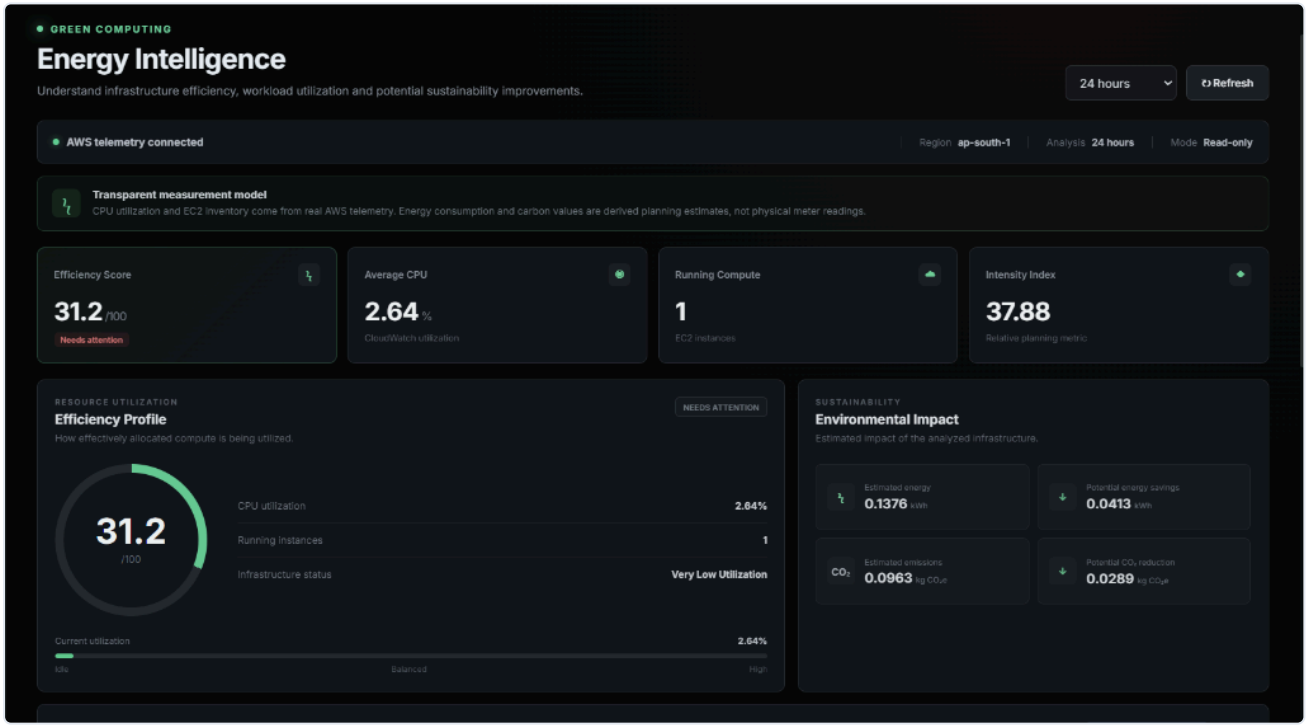


Figure 7.4: Energy Intelligence - Workload Utilization & Environmental Impact

Demonstrates live AWS telemetry connection in region `ap-south-1`, displaying continuous Efficiency Score (31.2/100), Average CPU (2.64%), and calculated sustainability metrics (0.1376 kWh energy, 0.0963 kg CO2e emissions).

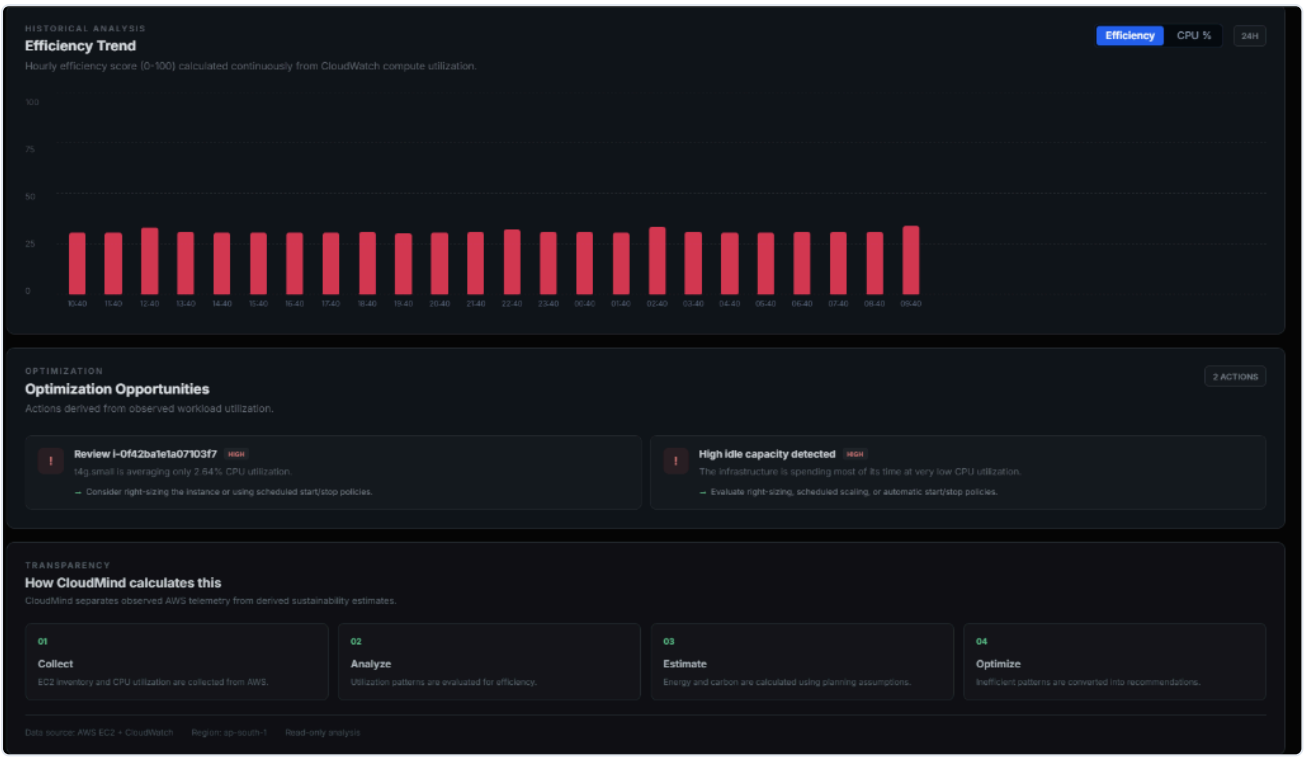


Figure 7.5: Historical Analysis - Continuous Efficiency Trend & Optimization Opportunities

Illustrates the resolved continuous telemetry trend chart across 24 hours (eliminating the flat 50% bug), rendering real dynamic fluctuations at ~30% along with automated right-sizing suggestions for instance `i-0f42ba1e1a07103f7`.

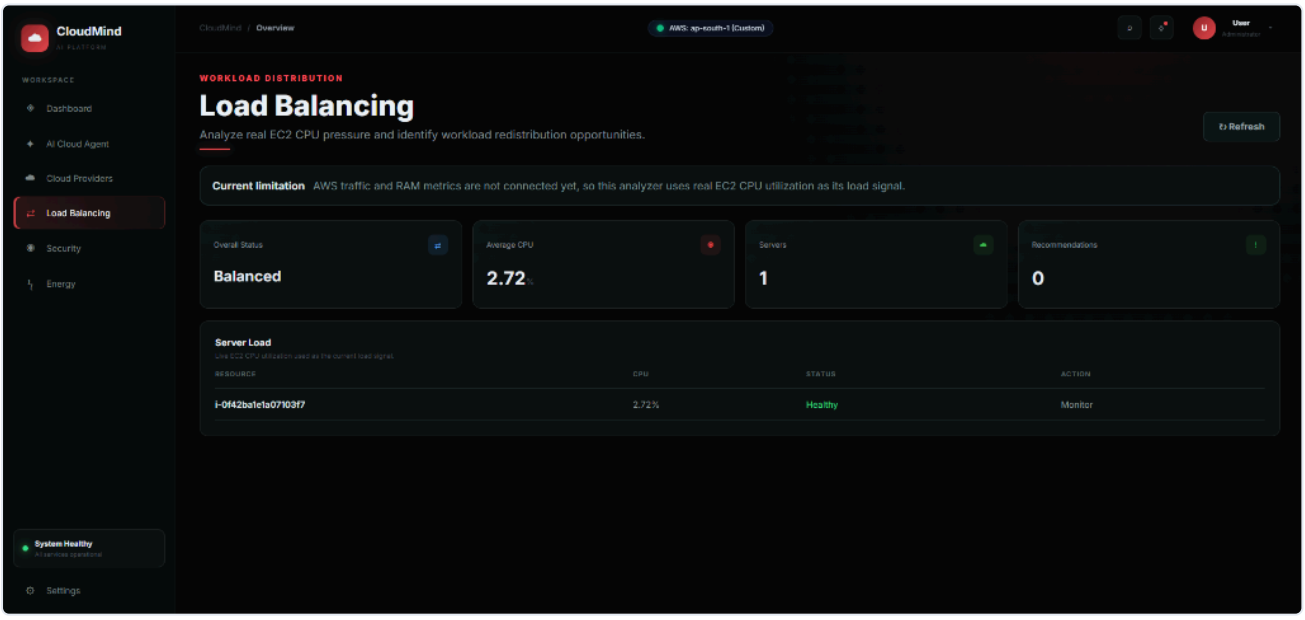


Figure 7.6: Workload Distribution - Real-Time EC2 Load Balancing & Server Health

Displays cluster status (Balanced), real-time CPU pressure (2.72%), server count, and active node health monitoring for compute instance `i-0f42ba1e1a07103f7`.

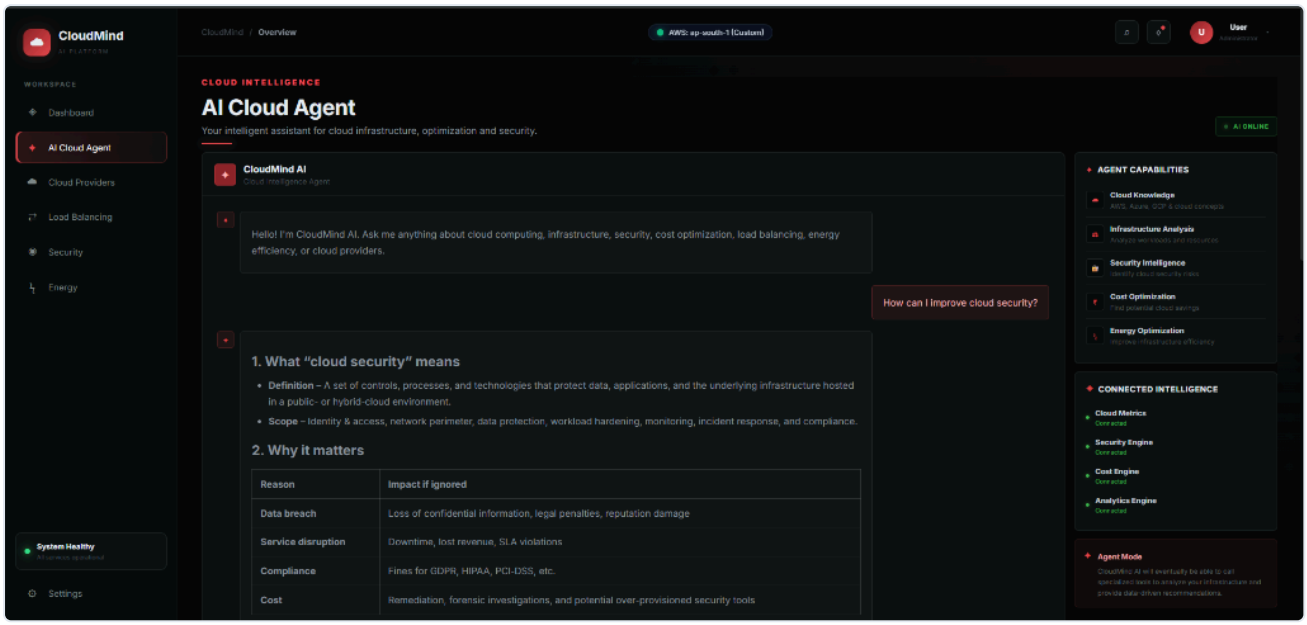


Figure 7.7: Autonomous AI Cloud Agent - Natural Language Infrastructure Copilot

Illustrates conversational cloud assistance answering security questions, risk assessments, and connected intelligence engines (Cloud Metrics, Security Engine, Cost Engine, Analytics Engine Connected).